iR4 / iR9 / iRDM20 / iRDM32 & Enhanced CDC-F

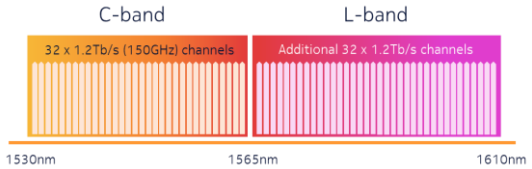
Product Overview

- v20 Oct ‘23
- Optical Networks

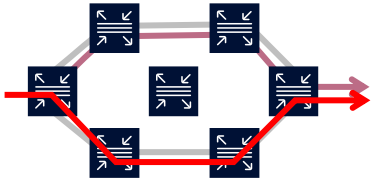
© Nokia 2023 | Confidential

Application-optimized optical line systems

Complete solutions to scale your critical network from edge to core







Scalable

- Double fiber capacity
- Modular and integrated C+L
- In-service upgrade/migration

Flexible

- Integrated ROADMs
- C-F and CDC-F options
- iR4/9/20/32 and WSS options

Resilient

- Proven network protection and restoration options
- Dynamic regeneration
- Maximize uptime and SLAs



Application optimized

- Compact modular DC-optimized chassis
- Integrated transponder/WDM telco-optimized chassis
- Range of chassis optimized from edge to core

© 2023 Nokia

Access/metro-optimized iROADMs

Delivering capacity, flexibility, and automation to the edge

iR4

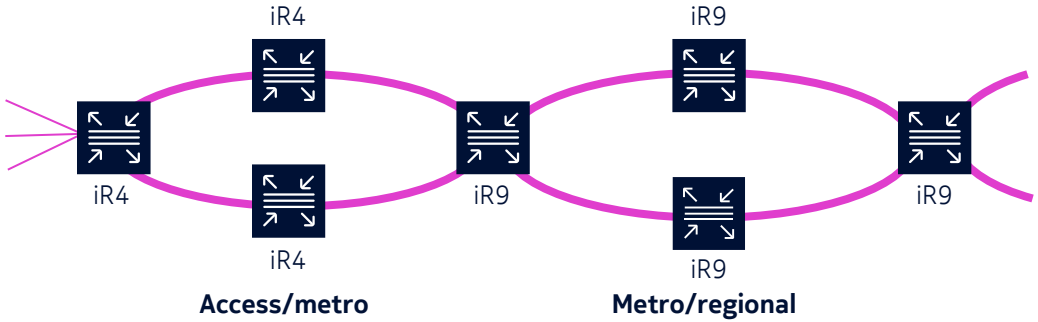
- Up to 4D ROADM
- Aggregation ring
- Metro edge/access
- C-F or mux/demux



iR9

- Up to 9D ROADM
- Metro/regional
- Ring-to-ring handoff





- ✓ Highly-integrated functionality
- ✓ Fixed and/or Flex Grid
- ✓ 1830 PSS or 1830 PSI platforms
- ✓ Common management



1830 PSS-8



1830 PSS-4II



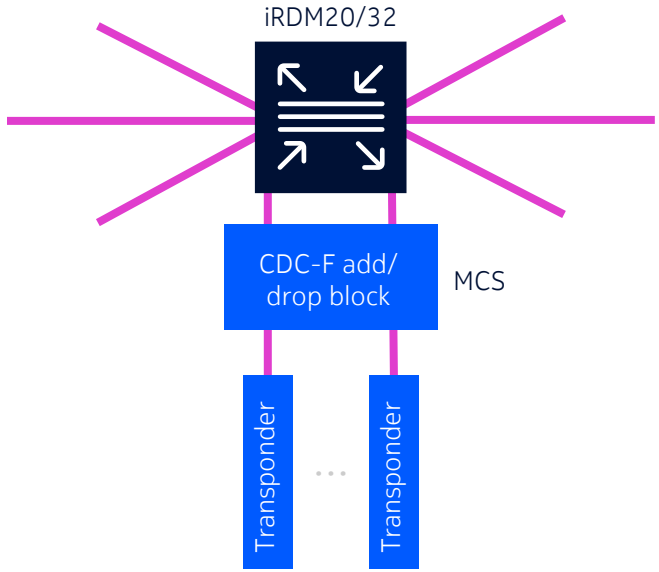
1830 PSI-4L

© 2024 Nokia

Core/backbone-optimized iROADMs

Delivering maximum scale and wavelength switching flexibility for core nodes

- Metro, regional, and long-haul applications
- FlexGrid options
- Up to 32 switching directions
- Up to 96 add/drop ports
- Up to 1,000 client-side transponder ports
- WSS options



iRDM20

- Up to 8D node sizes, full C-band



iRDM32

- Up to 16D node sizes, both C-band and L-band versions



© 2024 Nokia

Application-optimized chassis options

Common line system line cards into all chassis

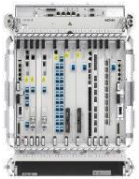
1830 PSS

- Mix and match line system and transponders
- 300mm depth for telco applications


1830 PSS-4II


1830 PSS-8


1830 PSS-16II


1830 PSS-32

1830 PSI-L

- Compact modular chassis
- Line system specific line-cards


1830 PSI-4L


1830 PSI-8L

Chassis options optimized for edge to core

© 2024 Nokia

Agenda



- IR4/iR9
- Nokia Enhanced CDC-F ROADM
- iRDM20/32
- C-F Configurations
- C+L Configurations
- PSI-L

Nokia Optical Networking Portfolio

- Versatile, Scalable, Dynamic
- Wide portfolio - WDM, OTN, Packet
- Capacity matched to application
- Seamless management

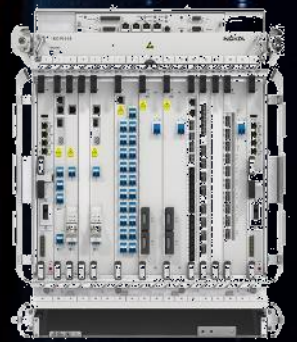












1830 PSI-M	1830 PSI-L	1830 PSS-8	PSS-16II	PSS-32

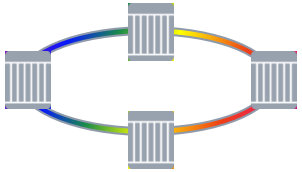
Optical Network Evolution – Flexibility, Capacity, Intelligence

Network Evolution

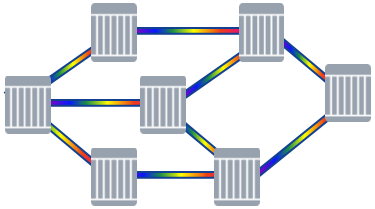
Pt – Pt Term

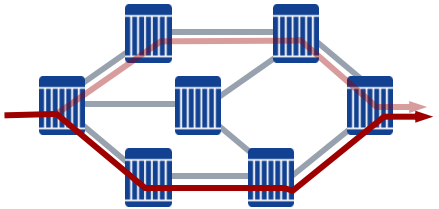


Ring Networks
- ROADM / WSS
- 10G wavelengths



Mesh Networks
- WSS ROADMs
- 100G Coherent
- OTN Switching / GMPLS (L1)



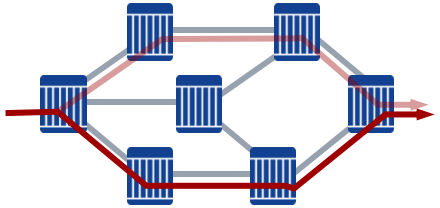


Dynamic Optical Networks
- CDC-F ROADMs
- 100G/200G Coherent
- GMPLS (L0/L1)
- Optical layer restoration / optimization

2005 2010 2015 2021

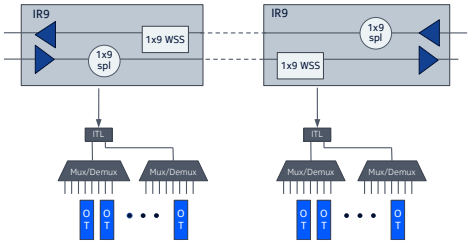
© Nokia 2023 | Confidential

Industry ROADM Evolution



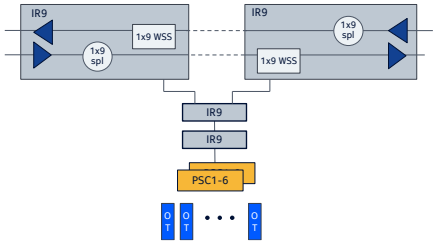
Nokia Supported ROADM Options

- Modern networks moving to CDC-F & C-F ROADMs



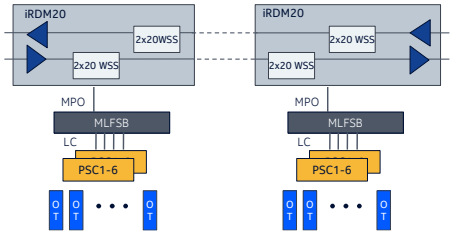
Classic ROADM

- Fixed ports
- Static wavelengths
- Most common in industry (until now)



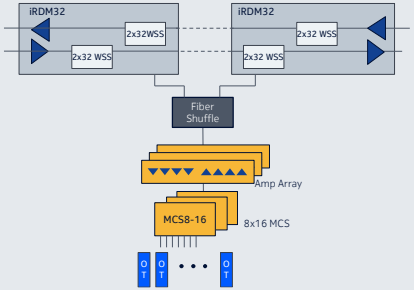
CD-F ROADM

- Dynamic - wavelength steering
- Colorless ports
- FlexGrid
- Ideal for legacy upgrades



C-F ROADM

- Colorless ports
- FlexGrid – enables new modulation types
- Static wavelengths

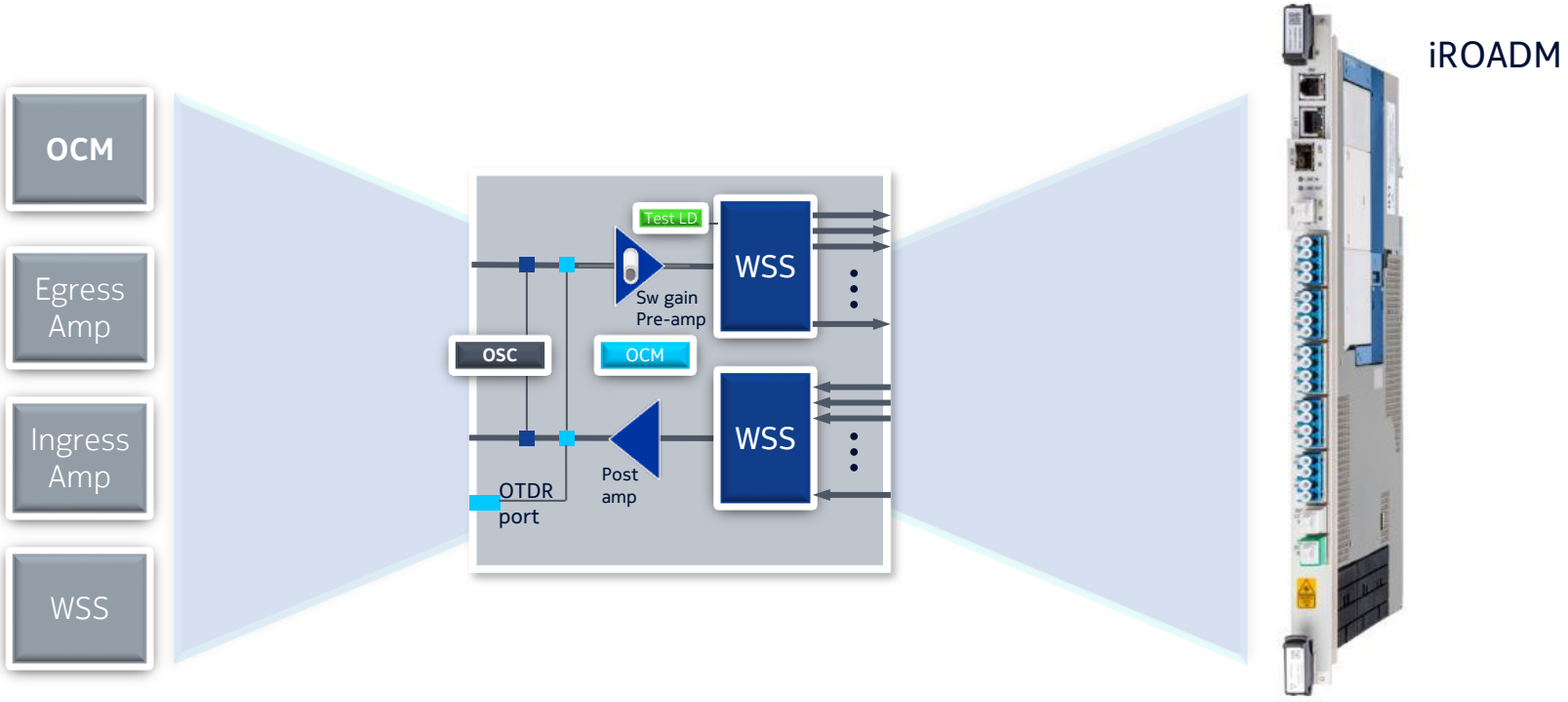


CDC-F ROADM

- Colorless, Directionless, Contentionless
- FlexGrid – enables new modulation types
- Dynamic
- Popular on modern core networks

© Nokia 2023 | Confidential

Integrated ROADM (iROADM)

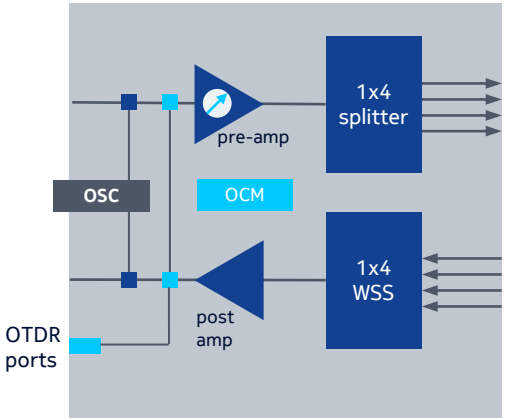


Continued innovation, integration for density & power improvements

 © Nokia 2023 | Confidential

iR4

Unit Name	iR4
Part #	3KC71462AA



Key Features:

- Optimized for 2D to 4D ROADM nodes
- Ideal for low cost access, metro, and regional applications
- Supports both classic ROADM (mux/demux) add/drop and FlexGrid add/drop (PCS1-6)
- 4800 GHz C-band operation
- Integrates variable gain pre-amp, post-amp, 1x4 WSS, 1x4 splitter, OCM into a single unit
- OTDR ports allows use of 1830 OTDR units (OTDR, OTDRM, ORTRWB)

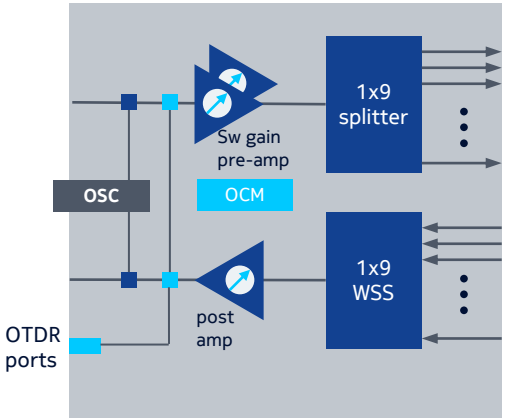


Unit Name	Type	Pre-amp Gain (dB) – Flat	Pre-amp Gain (dB) – Ext	Post-amp Gain (dB) - Flat	Pin min (dBm)	Pout max (dBm)	Avail
iR4	EDFA	15-25	21	20	-32	20	GA

© Nokia 2023 | Confidential

iR9

Unit Name	iR9 / iR9LP
Part #	3KC70869AA / 3KC70869AB



Key Features:

- Supports up to 9D ROADM nodes
- Classic ROADM (mux/demux) & FlexGrid add/drop
- Multi-haul WDM applications
- 4800 GHz C-Band
- Combines variable sw gain pre-amp, post-amp, WSS into single unit
- Updated version of iROADM9R
- “LP” version is lower output power (20 dBm)
  - Typically used in subsea applications (without OSC based APR)
- Span distances 0 – 32 dB



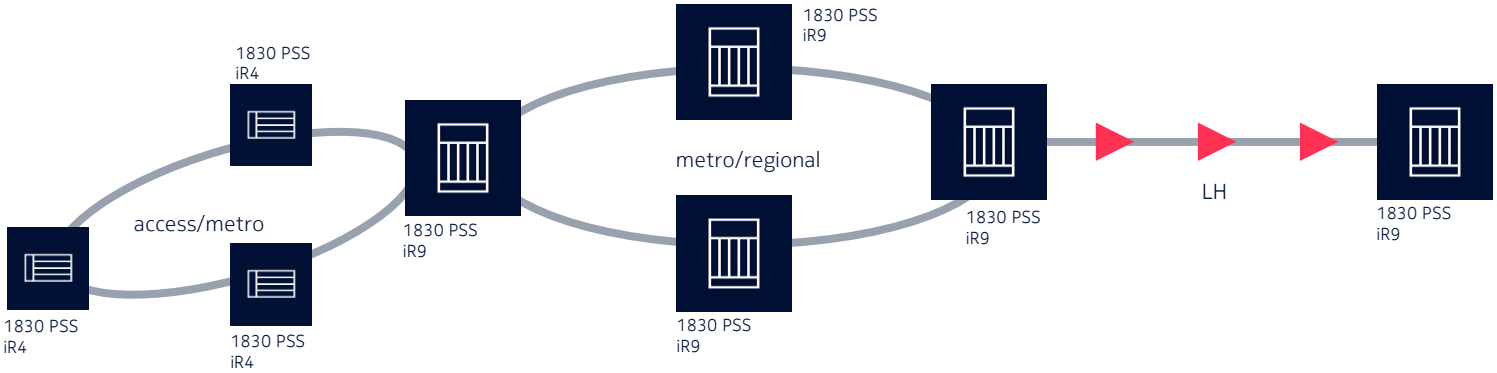
Unit Name	Type	Pre-amp Gain (dB) – Flat (low/hi)	Pre-amp Gain (dB) – Ext (low/hi)	Post-amp Gain (dB) - Flat	Post-amp Gain (dB) - Ext	Pin min (dBm)	Pout max (dBm)	Avail
iR9	EDFA	11-20 / 17-27	26/33	17-23	25	-34	23	GA
iR9LP	EDFA	11-20 / 17-27	26/33	17-23	25	-34	20	GA

© Nokia 2023 | Confidential

Applications

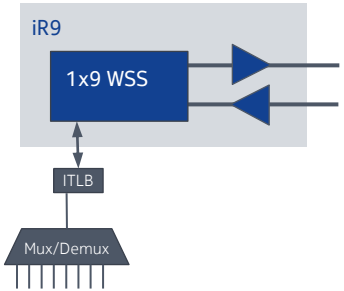
Ideal for:

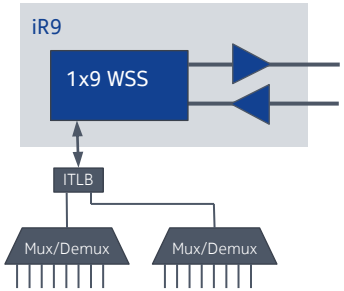
- Metro, regional, LH/ULH applications
- Same optical reach as existing WR88 + AM2xxx amps
- Supports both Mux/Demux (SFD44) & FlexGrid (PSC1-6) add/drop architectures
- Smaller size, lower cost replacement for existing WR88 + amps

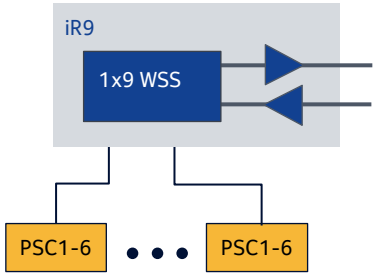


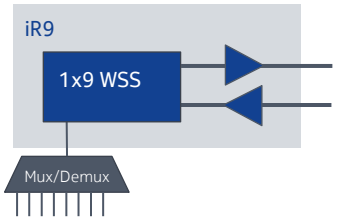
 © Nokia 2023 | Confidential

iR4 / iR9 Configurations









iR4 & iR9 Configurations

“Classic” ROADM

- Based on standard Broadcast & Select WSS architecture
- AWG Mux / Demux supporting:
  - 44 channels (w or w/o ITLB)
  - 88 channels
- Standard 50 GHz or 100 GHz channels
- Drop any wavelength at any ROADM node

Colorless FlexGrid ROADM

- Optimized for 100G & higher coherent
- ROADM architecture
- 1x6 Splitter / Couplers
- Colorless (any transponder to any port), Flexible Grid

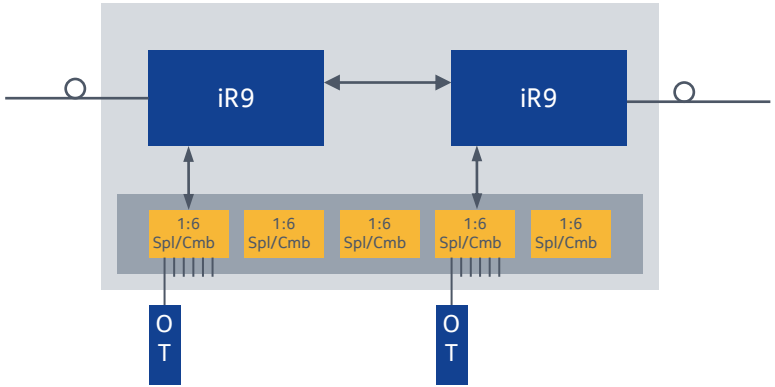
© Nokia 2023 | Confidential

iR4 / iR9 Colorless FlexGrid

Application

- 100G / 200G colorless, FlexGrid
- 100G & above applications
- Eliminates the need for SFD44/SFD44B Mux/Demux units
- PSC1-6 Splitter / Combiner
  - Incorporates (5) independent 1:6 Splitter / Combiner
  - Single shelf can support up to 30 coherent optical transponders
  - FlexGrid or fixed 50 GHz grid operation

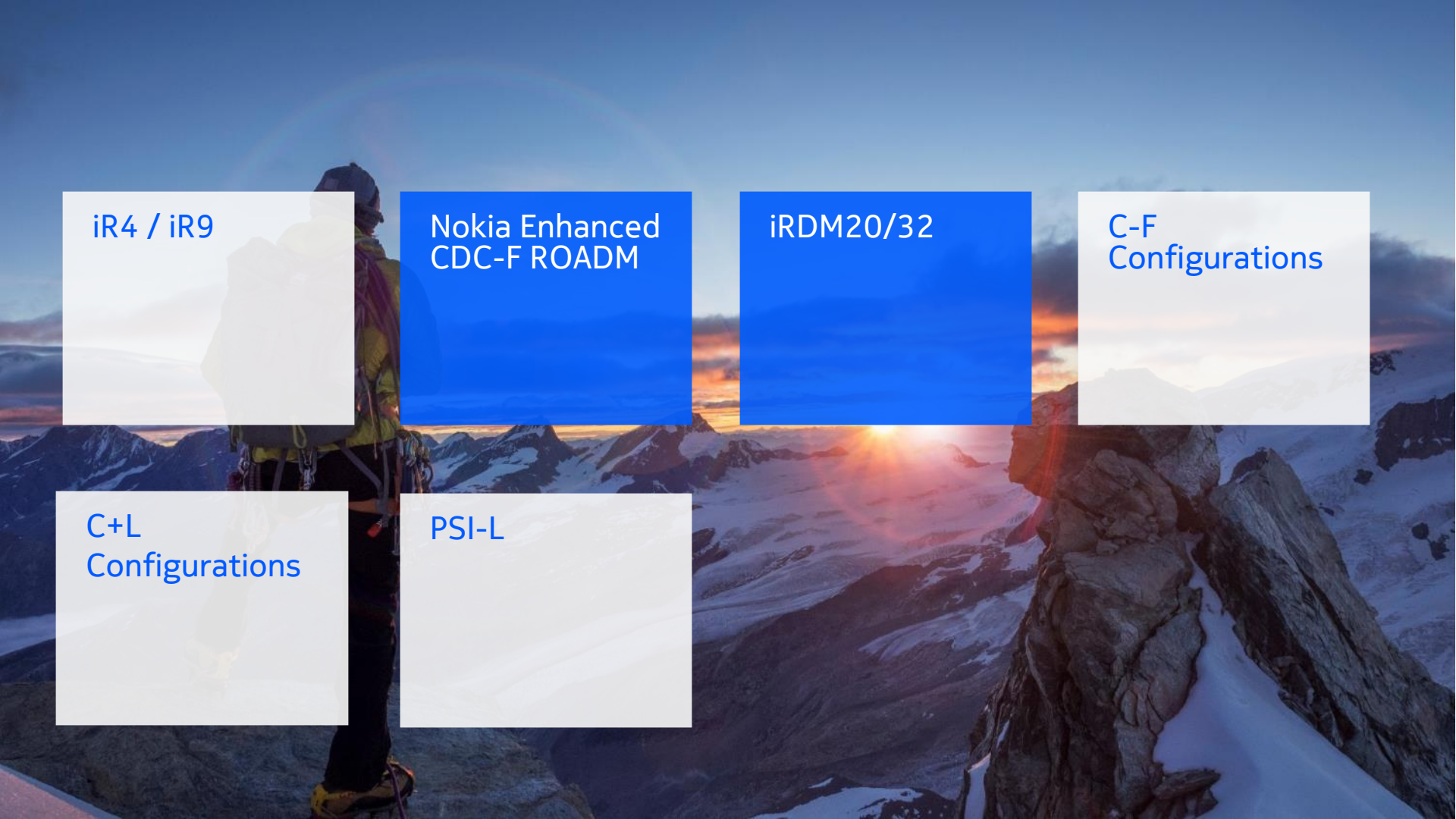
C-F ROADM node, 2D



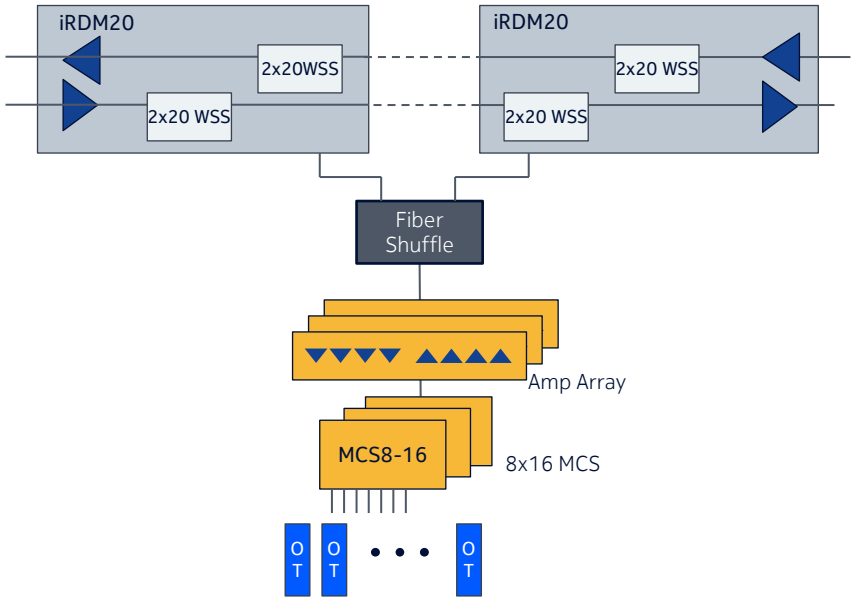
PSC1-6 Splitter / Combiner



© Nokia 2023 | Confidential


iR4 / iR9
Nokia Enhanced CDC-F ROADM
iRDM20/32
C-F Configurations
C+L Configurations
PSI-L

Nokia 1830 CDC-F ROADM



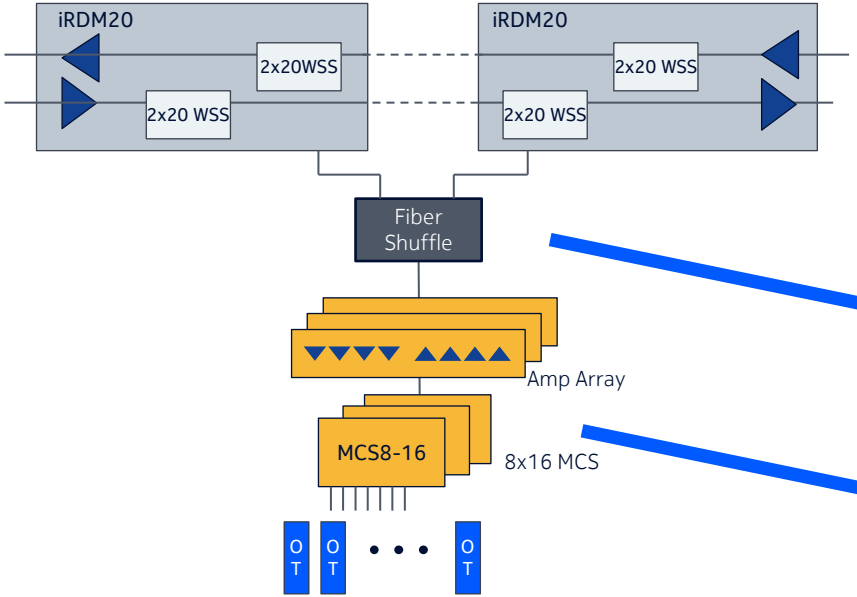
Note: 2D ROADM shown in diagram

Nokia 1830 CDC-F ROADM

- Switch / Steer wavelengths to any degree
- Colorless Ports
- Contentionless – no blocking
- Recover 30 – 70% additional network capacity
- Reduce ops costs - eliminate manual config.
- Support higher capacity with FlexGrid

© Nokia 2023 | Confidential

Nokia Enhanced (Gen 2.0) CDC-F ROADM



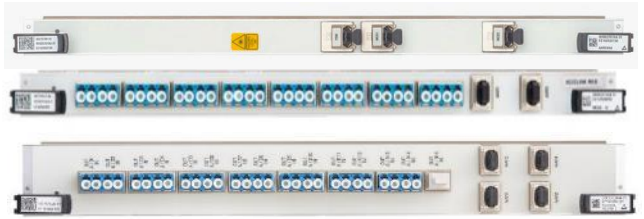


Integrated ROADM (iROADM):

- iRDM20, iRDM32
- Ingress, egress amps, OCM, OTDR ports



Modular Fiber Shuffle



CDC-F drop units:

- Amp Array
- 8x16 MCS
- 16x15 MCS

Note: 2D ROADM shown in diagram

© Nokia 2023 | Confidential

Nokia 1830 Enhanced CDC-F ROADM

Benefits

- 4x Density Improvement ideal for metro
- Support 16 deg nodes 2x Improvement
- 4x capacity Up to 768 client ports (192 Tb)

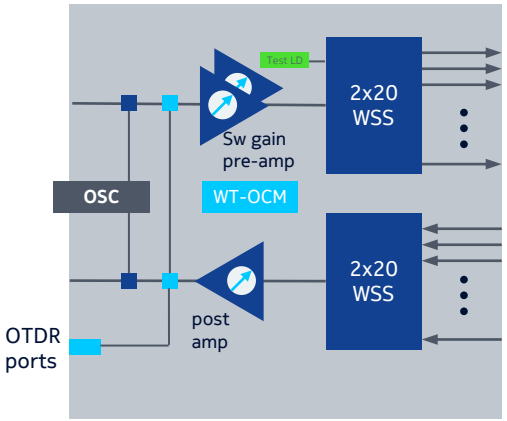
Technology Advantages

- Integrated ROADM Single Card
- CDC-F ROADM Optical Layer Flexibility
- Full 4800 GHz C-Band

© Nokia 2023 | Confidential

iRDM20 – Twin WSS Integrated ROADM

Unit Name	IRDM20
Part #	3KC69751AA
Size	Full Height, 1-slot





iRDM20

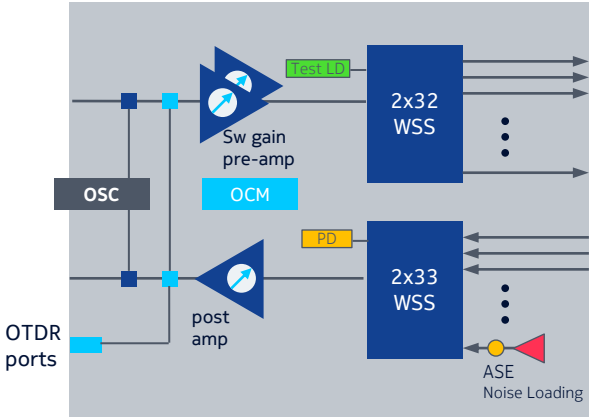
- iRDM20 - Twin 2x20 WSS with Sw gain amp
- Ideal for CDC-F applications
- Significant size & cost reduction
  - Support up to 4D (CDC-F) node in single 1830 PSS-16II shelf
- 96 channel capable (4800 Hz)
- Flexible Grid
- Integrated test signal
- New, modular fiber shuttle panel (MSH4-FSB) simplifies fiber connections

Unit Name	Part #	# Deg	# Chan	Pre-Amp / Post – Amp	Span Range (approx)	Size (H / W)	Supported Chassis	Avail
iRDM20	3KC69751AA	≤12	96	Sw Gain Var/ Fixed	0 – 30 dB	Full / 1-slot	PSS-16II, -32	GA

© Nokia 2023 | Confidential

iRDM32 – Twin WSS Integrated ROADM

Unit Name	IRDM32 / IRDM32L / iRDM32LP
Part #	3KC70163AA / 3KC70342AA / 3KC71158AA
Size	Full Height, 1-slot





iRDM32

- iRDM32 - Twin 2x32 WSS with switched gain pre-amp
- Ideal for CDC-F / C-F applications
- “LP” version is limited output power (20 dBm)
  - Typically used in subsea applications (w/o OSC based APR)
- 4800 Hz spectrum
- Flexible Grid
- Integrated test signal
- Integrated OCM

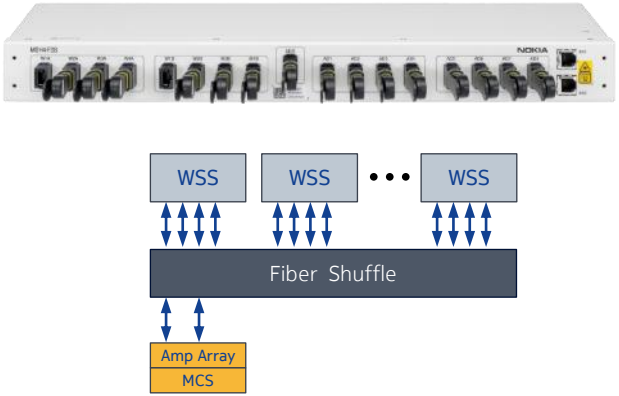
Unit Name	Part #	Pre-Amp Lo Gain (dB)	Pre-Amp Hi Gain (dB)	Post – Amp (dB)	Post Amp Power - max (dBm)	Supported Chassis	Avail
iRDM32	3KC70163AA	10 – 24	16 - 32	18 -25	24	PSS-8, -16II, -32, PSI-L	GA
iRDM32L	3KC70342AA	10 – 24	16 - 32	18 -25	24	PSS-8, -16II, -32, PSI-L	GA
iRDM32LP	3KC71158AA	10 – 24	16 - 32	18 -25	20	PSS-8, -16II, -32, PSI-L	GA

© Nokia 2023 | Confidential

MSH4-FSB Fiber Shuffle

MSH4-FSB

- Modular, bidirectional, fiber shuffle (ie fiber interconnection panel)
- High capacity 12-fiber MPO cables simplify intra-node connections between units
  - Eliminates need for hundreds of individual fiber connections
- Provides in trade-off between using WSS ports for larger degree sizes vs drop capacity
- Modular – add additional fiber shuffle units as needed by node size and/or drop capacity

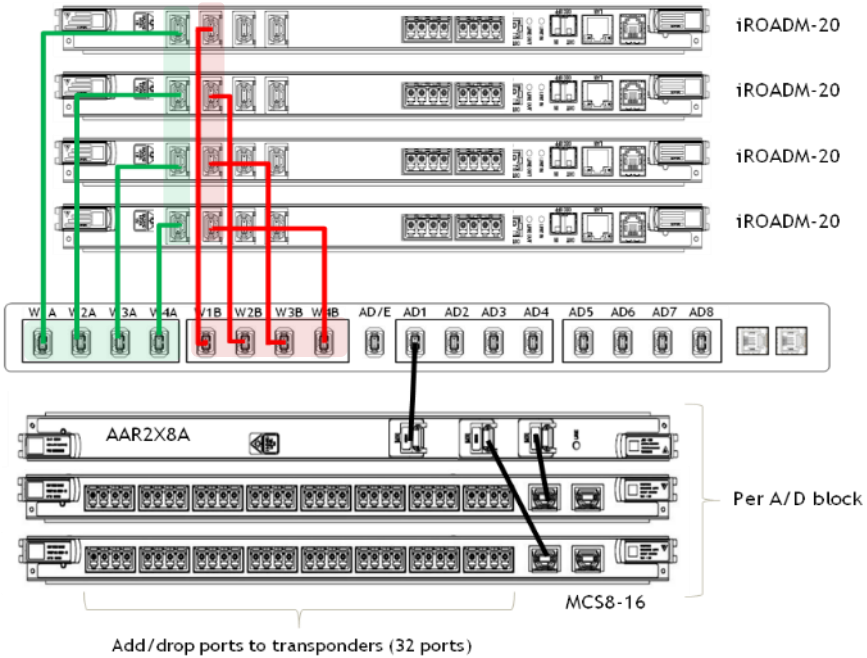


Degree	1x20 WSS (iRDM20)					1x32 WSS (iRDM32)				
	# Avail ROADM ports	Trpn Ports 8x16 MCS	% Total A/D Capacity	Trpn Port 16x15MCS	% Total A/D Capacity	# Avail ROADM ports	Trpn Ports 8x16 MCS	% Total A/D Capacity	Trpn Port 16x15MCS	% Total A/D Capacity
1-4	17	192	100%	192	100%	29	192	100%	192	100%
5-8	13	384	100%	384	100%	25	384	100%	384	100%
9-12	9	n/a		270	47%	21	n/a		576	100%
13-16	5	n/a		150	20%	17	n/a		510	66%

Note: based on 48-chan / 100 GHz wavelengths

© Nokia 2023 | Confidential

4 Deg Node Interconnection Example



MSH4-FSB Fiber Shuffle

- Simplified intra-node fiber connections
- 12 fiber MPO cables

© Nokia 2023 | Confidential

8 Deg Node Example

Degree 1-4



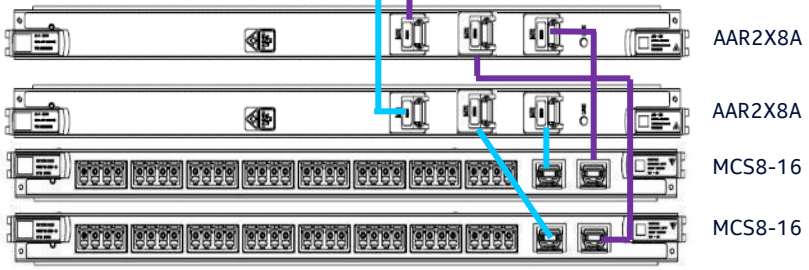
Degree 5-8





Deg 1-4 drop

Deg 5–8 drop



8D Connection

- (4) MSH4-FSB units
- 2nd set of red connections (Deg 5-8) moved to first set of fiber shuffle unit to proved node interconnections

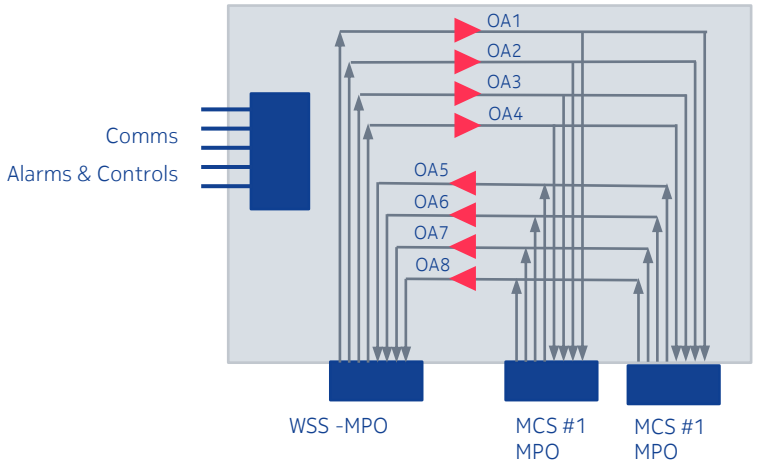
 © Nokia 2023 | Confidential

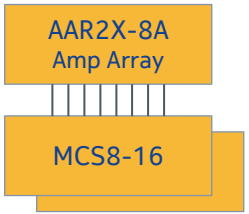
AAR2X-8A Amp Array

Unit Name	AAR2X-8A
Part #	8DG63767AA / 8DG63768AA
Size	Full Height, 1-slot

Key Features:

- Supports double drop capacity (2x)
- Each AAR2X-8A supports up to (2) MCS8-16 Multicast Switch units
- Up to 32 drop ports total
- Amp Array to compensate for optical losses on CDC-F drop units (MCS)
- Incorporates (8) amps, (4) Tx & (4) Rcv
- Provides MPO connections to Twin 1x20 WSS and 8x16 MCS





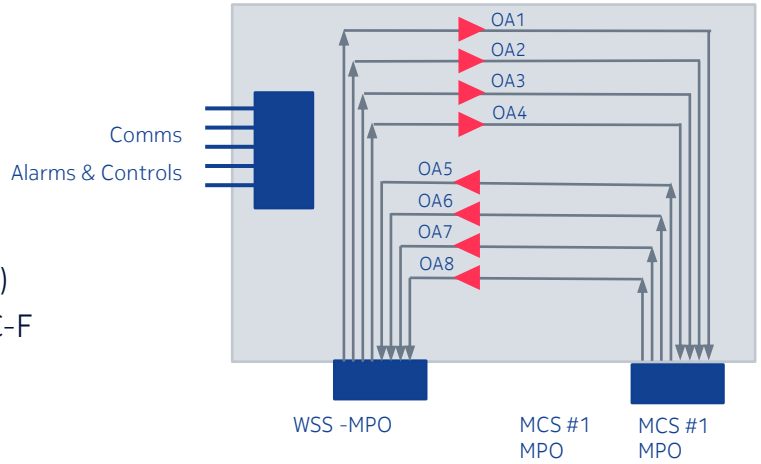


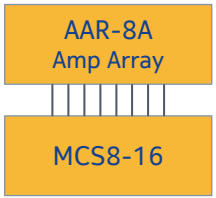
AAR2X-8A

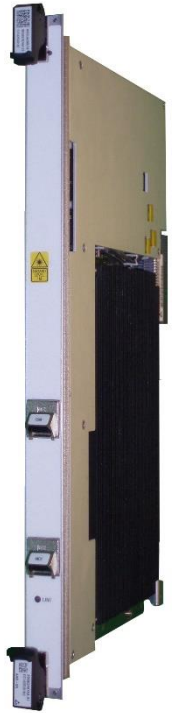
 © Nokia 2023 | Confidential

AAR-8A Amp Array

Unit Name	AAR-8A / AARL
Part #	8DG62497AA / 3KC71973AA
Size	Full Height, 1-slot







AAR-8A

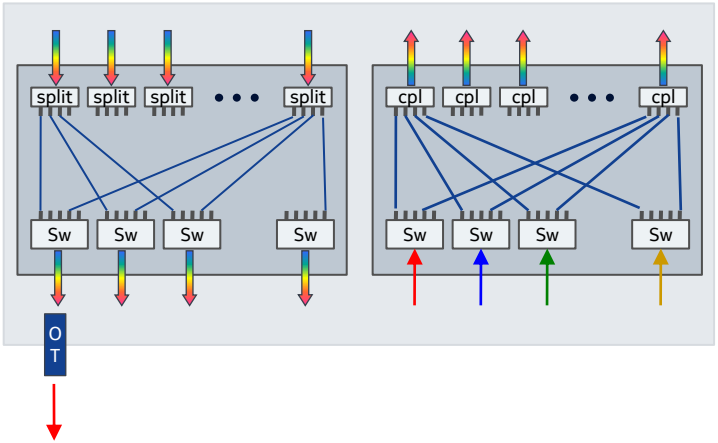
Key Features:

- Amp Array used with multi-cast switch cards (MCS)
- Amp Array compensates for optical losses on CDC-F drop units (MCS)
- Each AAR-8A supports single MCS unit
  - Slightly better OSNR performance with single MCS card
- Up to 16 drop ports total
- Incorporates (8) amps, (4) Tx & (4) Rcv
- Provides MPO connections to Twin 1x20 WSS and MCS cards
- L-Band version (AARL)

© Nokia 2023 | Confidential

MCS8-16 Multicast Switch

Unit Name	MCS8-16 / MCS8-16L
Part #	8DG62474AA / 8DG63101AA
Size	Full Height, 1-slot





MCS8-16

Key Features:

- Technology enabling CDC-F ROADMs
- 8 x 16 MCS
  - Connects up to 8 Degrees
  - Supports (16) drop ports
- Contentionless – can drop same color wavelength
- Operation - Drop
  - Incoming signals split to each drop port (16)
  - 1x8 switch on the drop port, provisioned to a specific degree
  - Coherent transponder “tunes” to the provisioned channel
- Operation - Add
  - 1x8 switch connects client port to one of (8) network degrees
  - Combiner adds signals from all 16 client ports
- Requires use of AAR-A8 Amp Array due to high insertion loss (17 dB max)

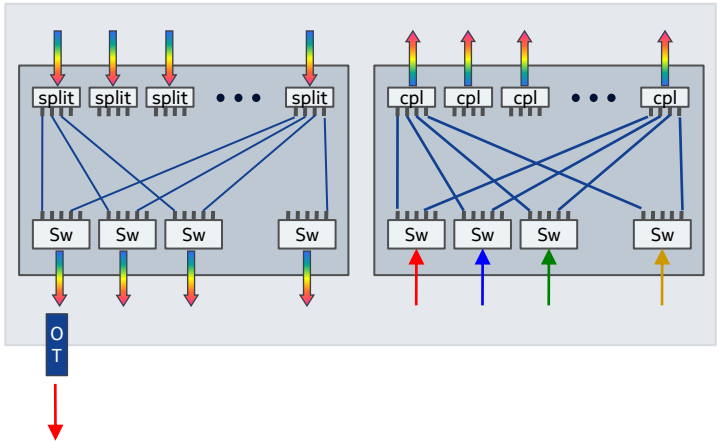
© Nokia 2023 | Confidential

MCS16-15 Multicast Switch

Unit Name	MCS16-15 / MCS16-15L
Part #	3KC70234AA / 3KC70514AA
Size	Full Height, 2-slot

Key Features:

- Technology enabling CDC-F ROADMs
- 16 x 15 MCS
  - Connects up to 16 Degrees
  - Supports up to 30 drop ports (via 2 units), per amp array
- Contentionless – can drop same color wavelength
- Operation - Drop
  - Incoming signals split to each drop port (15)
  - 1x15 switch on the drop port, provisioned to a specific degree
  - Coherent transponder “tunes” to the provisioned channel
- Operation - Add
  - 1x15 switch connects client port to one of (16) network degrees
  - Combiner adds signals from all 15 client ports
- Integrated test laser / power detector for automated connection verification



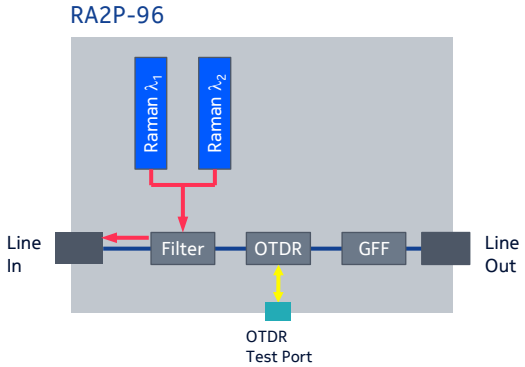


MCS16-15

 © Nokia 2023 | Confidential

RA2P-96 Raman Amp (optional)

Unit Name	RA2P-96
Part #	8DG64137AA
Size	Full Height, 1-slot





RA2P-96

Key Features:

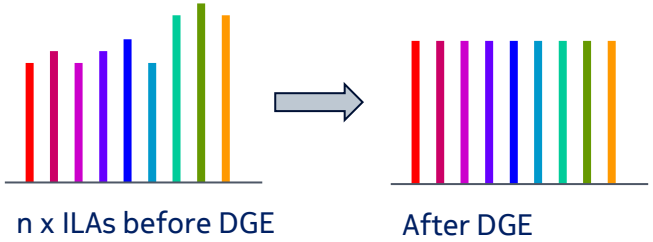
- New, enhanced RA2P-96 supporting up to 96 channels C-band
- Integrated OTDR ports
- Intended for LH applications or long span applications that require additional boost
- Includes a gain flattening filter designed to deliver optimum flatness
- Two pumps (1425 nm / 1454 nm) - 760 mW total power
- Net target gain of the DWDM channels: 11 dB for SSMF; 15 dB for LEAF; 17 dB for TW

© Nokia 2023 | Confidential

E-ILA Dynamic Gain Equalization (DGE)

Dynamic Gain Equalization (DGE)

- Provides auto-power balancing across the wavelengths
- Corrects for EDFA gain tilt
- Typically required every 5 or 6 ILAs on long haul routes



E-ILA Node

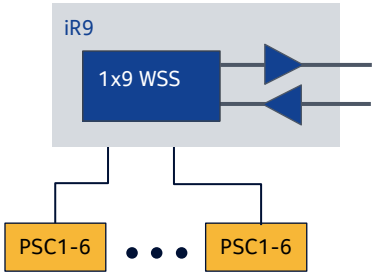
- Equalized ILA = East/West amps + OCM + DGE (single card)
- Smaller, updated version of existing DGE solution
- Available in C-band (EILA), L-band (EILAL), & C+L (OMDCL+EILA+EILAL) configs



© Nokia 2023 | Confidential


iR4 / iR9
Nokia Enhanced CDC-F ROADM
iRDM20/32
C-F Configurations
C+L Configurations
PSI-L

iRDM20/32 C-F Configurations

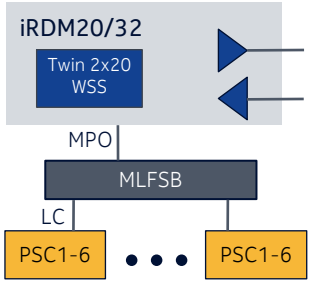




PSC1-6 Splitter / Combiner

iROADM Colorless FlexGrid (C-F) ROADM

- iR9 – Available Now
  - Colorless (any transponder to any port), FlexGrid
  - Optimized for 100G & higher coherent
  - iR9 & 1x6 Splitter / Couplers



- iRDM20/32 C-F Configurations
  - iRDM20/32
  - Colorless – FlexGrid (C-F)
  - FlexGrid enables use of newer 200G – 600G wavelength modes requiring new channel spacings (62.5 GHz, 75 GHz, 87.5 GHz, etc)
  - Smaller size, lower cost (C-F) option – where full CDC-F ROADM isn't required (ie Pt – Pt)
  - MLFSB provides MPO – LC breakout for connection to PSC1-6

© Nokia 2023 | Confidential

MPO-LC fiber shuffle (MLFSB)

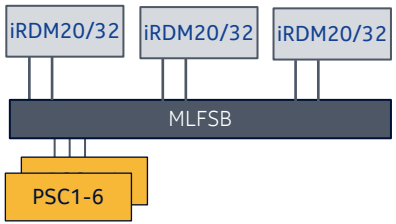
Unit Name	MLFSB
Part #	3KC70692AA

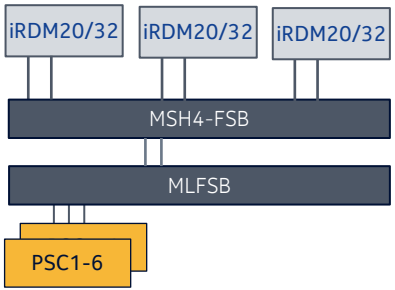


Key Features:

- Bidirectional MPO-LC fiber shuffle
- High capacity 12-fiber MPO cables to LC break out
- 6 MPO-LC modules in each shelf
- (1) RU
- Provides direct connection to iRDM20/32 MPO ports
- Optional use with MSH4-FSB for mixed C-F / CDC-F applications

Degree	iRDM20	
	# Add/drop ports/ degree	A/D channels PSC1-6
1	20	120
4	17	102
8	13	78
12	9	54
16	5	30





Degree	iRDM32	
	# Add/drop ports/ degree	A/D channels PSC1-6
1	32	192
4	29	174
8	25	150
12	21	126
16	17	102

 © Nokia 2023 | Confidential

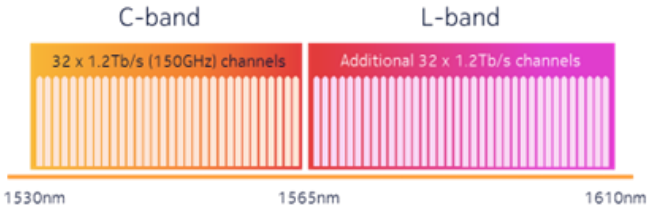

iR4 / iR9
Nokia Enhanced CDC-F ROADM
iRDM20/32
C-F Configurations
C+L Configurations
C+L Configurations

Behind the bands

C-band and L-band WDM

C-band
- Conventional band
- 1530nm – 1565nm
- Lowest loss wavelength band
- Spectrum has evolved to 4,800GHz (4.8THz)

L-band
- Long band
- 1565nm – 1625nm
- 2nd lowest loss
- L-band only systems for certain fiber types

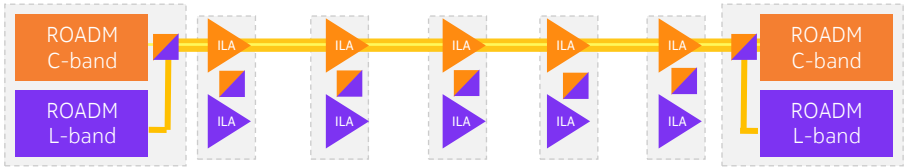


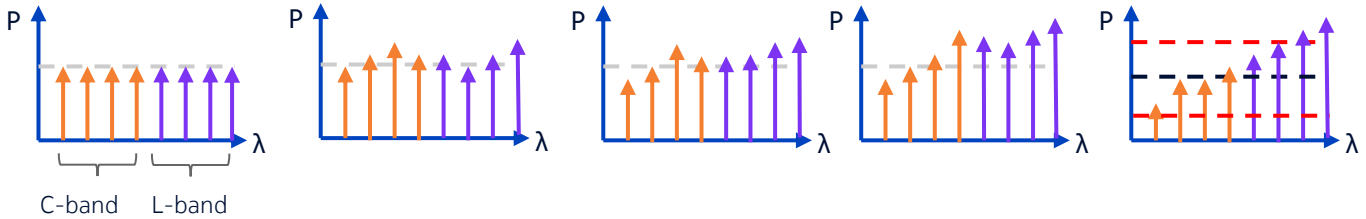
C + L
- Doubles available spectrum to 9,600GHz
- “Super” C and L solutions provide up to 12,000GHz
- Stimulated Raman Scattering (SRS) can affect network performance and wavelength management

© 2023 Nokia

Understanding Stimulated Raman Scattering (SRS)

- Non-linear effect that causes a power transfer from shorter wavelengths to longer wavelengths
- Causes wavelength “tilt” across C and L bands
- Impacts design, provisioning, and restoration



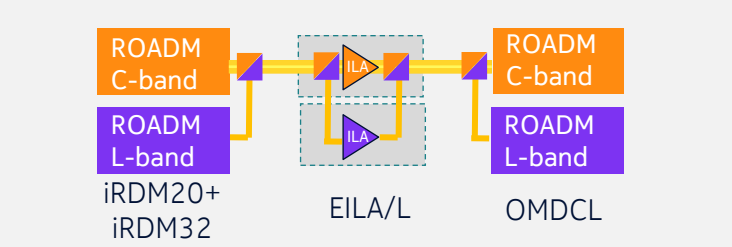


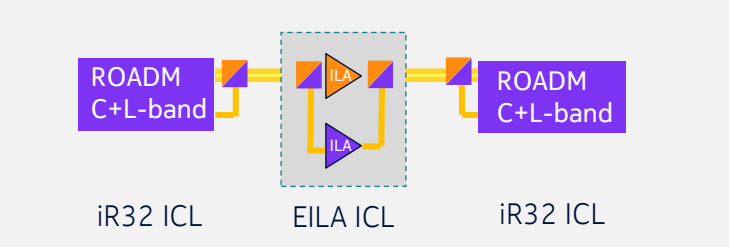
Nokia 1830 C+L systems automatically monitor & adjust SRS Tilt

© 2023 Nokia

C+L solutions

Full suite of C+L solutions optimized to different deployment scenarios





Modular C+L

- Modular C- and L- ROADMs
- Single-card EILAs/DGE
- OMDCL C+L coupler
- Seamless scalability with maximum flexibility

Integrated C+L

- Fully integrated C+L
- Integrated C+L iROADM
- Integrated C+L EILA
- Maximum density and operational simplicity

© 2024 Nokia

Seamlessly scale fiber capacity as needed

Modular C+L solution – scalability your way



Modular C+L solution

- L-band ready, pay-as-you-grow solution doubles network capacity
- Up to 9.6 THz capacity
- Cost-effective L-band upgrade options
  - Pre-deploy L-band with band couplers, amps for faster upgrade and lower operational costs, or…
  - Defer until needed, PAYG for low initial equipment costs
- Automatic self-tuning for SRS tilt across C+L

© 2024 Nokia

Seamlessly upgrade to protect network investments

Progressive C+L solution enables mixed node upgrades

Network-level scaling

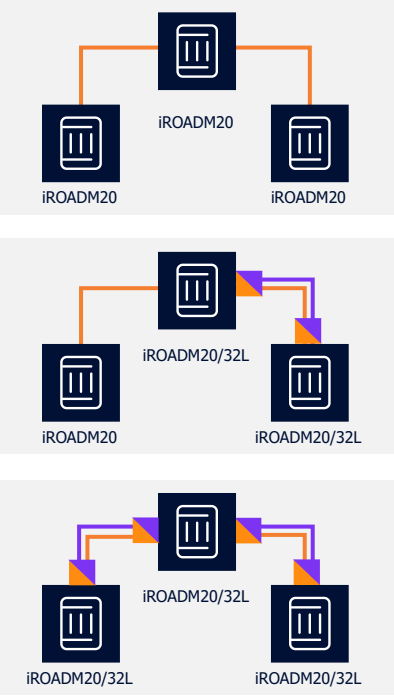
- Pay-as-your-network upgrades
- L-band upgrades double network capacity
- Avoids the cost of additional fiber pairs

Link-level scaling

- Pay as your traffic grows link-by-link
- No need to uniformly upgrade all links
- Upgrade adjacent links for C+L node islands

ROADM-level scaling

- Pay as your add/drop degrees grow



C-band only CDC-F
add / drop C-band traffic

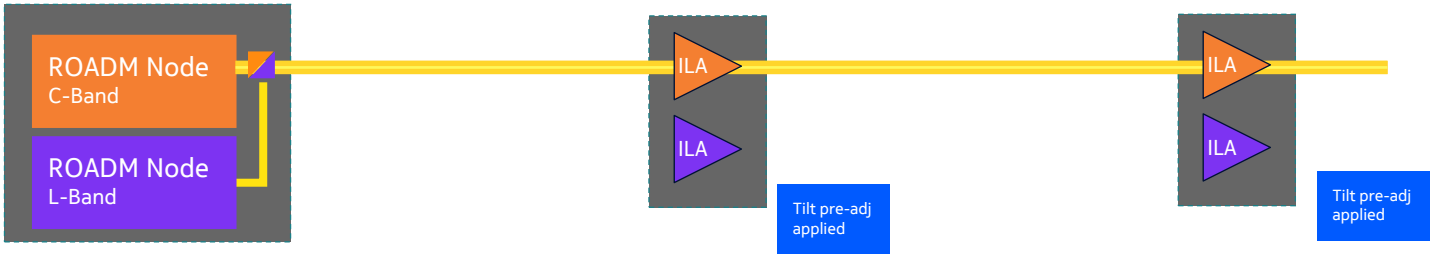
Selective node/link upgrades
Add L-band

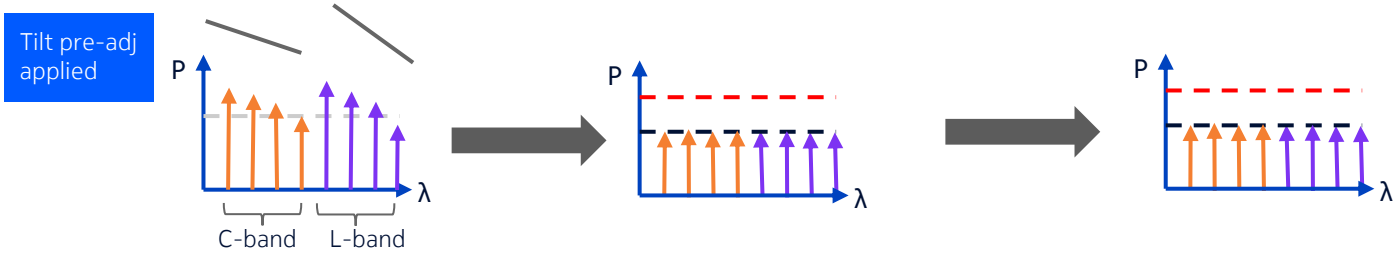
Full L-band upgrades
Add/drop C & L band all nodes

© 2024 Nokia

Nokia self tuning C+L networks – how it works

- 1830 C+L system applies tilt pre-adjustment profile to each band
- The “tilt pre-adjustment” automatically updates as wavelengths are added or deleted





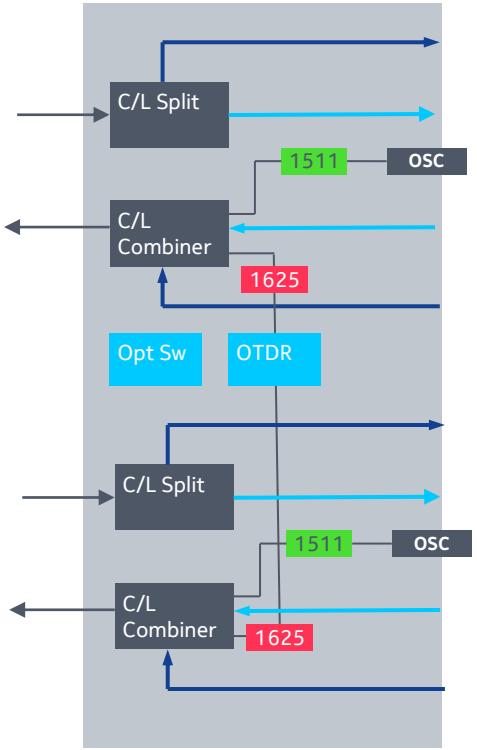
© 2023 Nokia

C+L Combiner / Splitter

Unit Name	OMDCL
Part #	3KC70424AA
Size	Full Height, 1-slot

- OMDCL
  - Separate C+L Combiner / Splitter
  - Allows for standalone C, L, or C+L operation
  - Integrated wideband OTDR





 © Nokia 2023 | Confidential

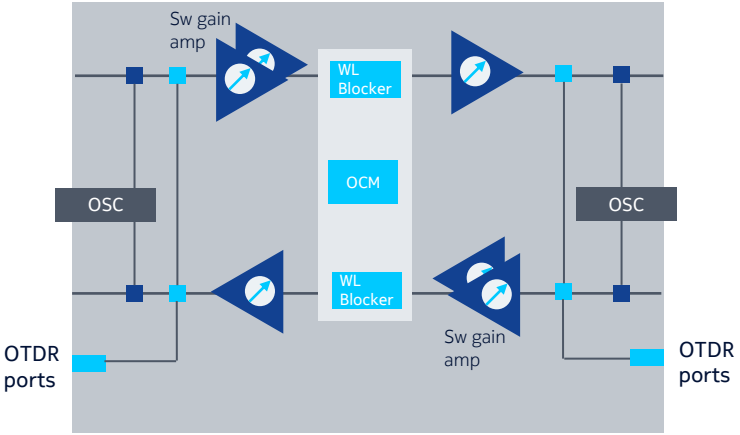
Equalized ILA C-Band & L-Band (EILA / EILAL)

Unit Name	EILA (C-band) / EILAL (L-band)
Part #	3KC70175AA / 3KC70525AA
Size	Full Height, 1-slot

EILA



	Features	
Features	• Equalized ILA – used in DGE node applications • Integrated Sw Gain Amps, wavelength blocker / OCM • Dynamic gain equalization manages wavelength powers • EILA (C-band) and EILAL (L-Band) versions • Integrated OTDR ports & MON ports (not shown)	
Operating Range	• C-Band • L-Band	1528.58 nm to 1567.7 nm 1570.21 nm to 1611.5 nm
Specs	• Power Out • Max gain • Blocker Atten • Chan BW	24 dBm 32 dB 0 to 15 dB 25 to 4900 GHz
Shelves	• PSS-8/16II/32 PSI-4L/8L	



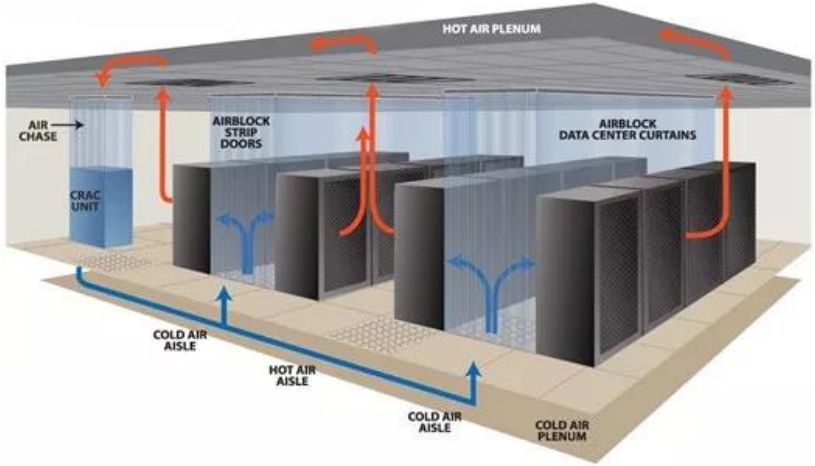
 © Nokia 2023 | Confidential

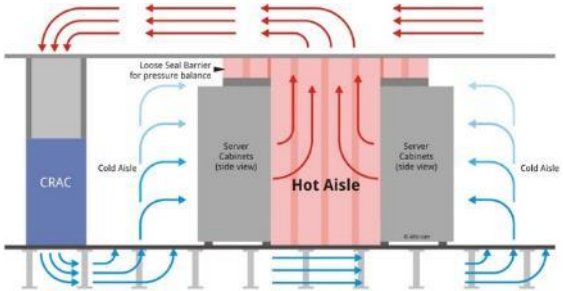

iR4 / iR9
Nokia Enhanced CDC-F ROADM
iRDM20/32
New C-F Configurations
C+L Configurations
New PSI-L Configurations

Industry Trends - Environment & Physical

DCI Form Factor

- Front to back airflow
  - Hot aisle / cold aisle layout (GR-3160)
  - Front / side exhaust is not allowed - disrupts optimal airflow / thermal design
- 19” rack / cabinet mounting
  - Data centers standardized on 19” racks / cabinets





Data centers use Hot Aisle / Cold Aisle concept (GR-3160)

Front to back airflow is a requirement

© Nokia 2023 | Confidential

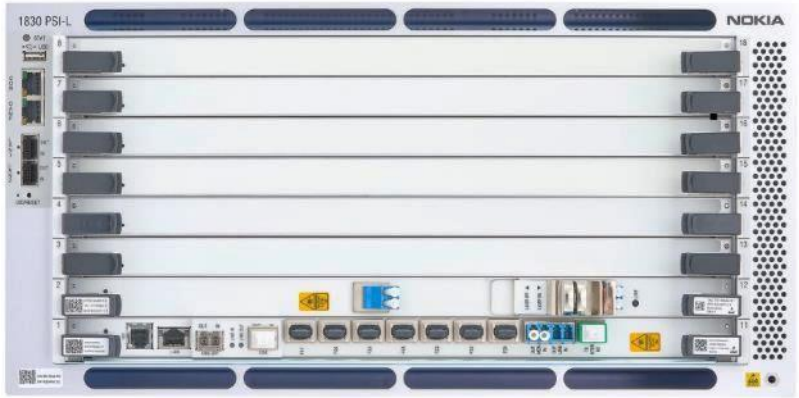
PSI-L

Nokia PSI-L Shelf Options

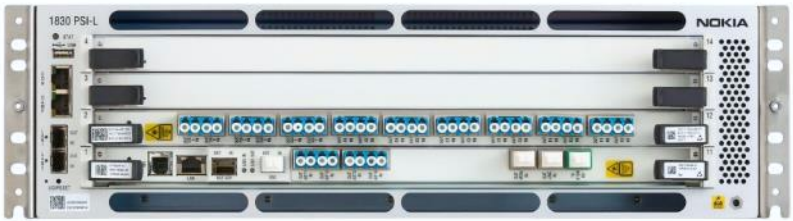
- New shelf options aligned with industry trends
  - Front – Back airflow
  - 19” / 23” rack or cabinet mount
  - AC or DC powered
  - Supports WDM line system cards only – no transponders
- Supports iROADM9/20/32 C-F, CDC-F configurations

- PSI-8L 8-slot shelf
  - Supports existing 1830 WDM line units
  - 5RU high, ~435mm depth, 19” / 23” mounting

- PSI-4L 4-slot shelf
- Essentially ½ capacity option for smaller configurations



8-slot



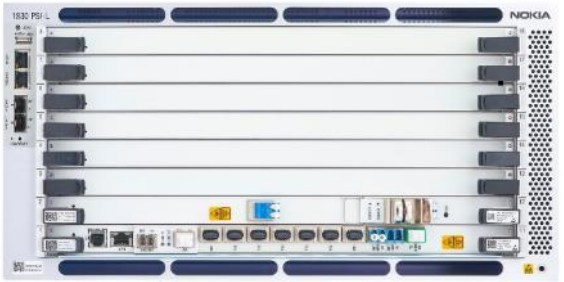
4-slot

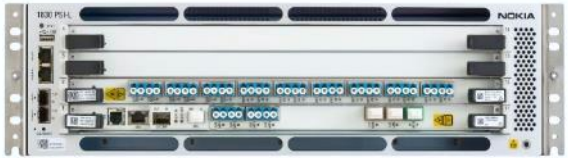
© Nokia 2023 | Confidential

Photonic Service Interconnect – Line

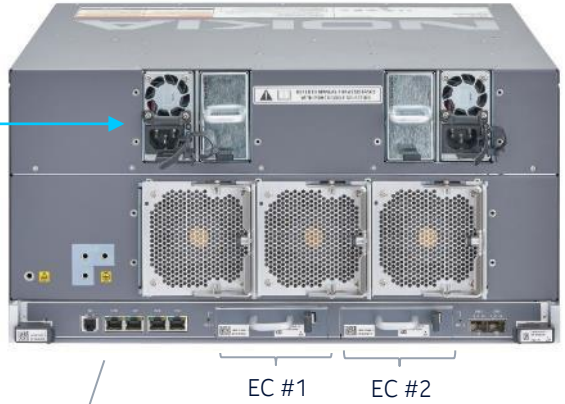
Compact modular photonic line system – DCI optimized

- Introducing Nokia 1830 PSI-L
- Front to rear airflow
  - 435mm depth – enables baffling / airflow through shelf
- Modular AC or DC power options
- Designed for 19” or 23” rack / cabinets



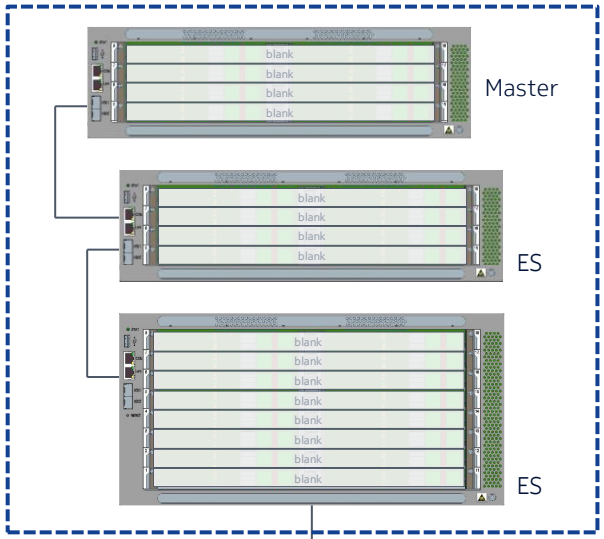






 © Nokia 2023 | Confidential

PSI-L ROADM Application





PSI-M



PSS-8/16II/32

PSI-L Multi-Shelf Configuration

- Introduction of PSI-L into standard PSS Rel 13.0+ software
- PSI-L becomes deployable for all customers – simply another shelf option to -8/-16II/32
- System supports mix of PSI-L 4-slot or 8-slot shelves
  - No mixing with legacy shelf types in the same node (ie -8/16II/32)
- Transponders
  - PSI-M or PSS-32/16II/8 in “cluster” configuration

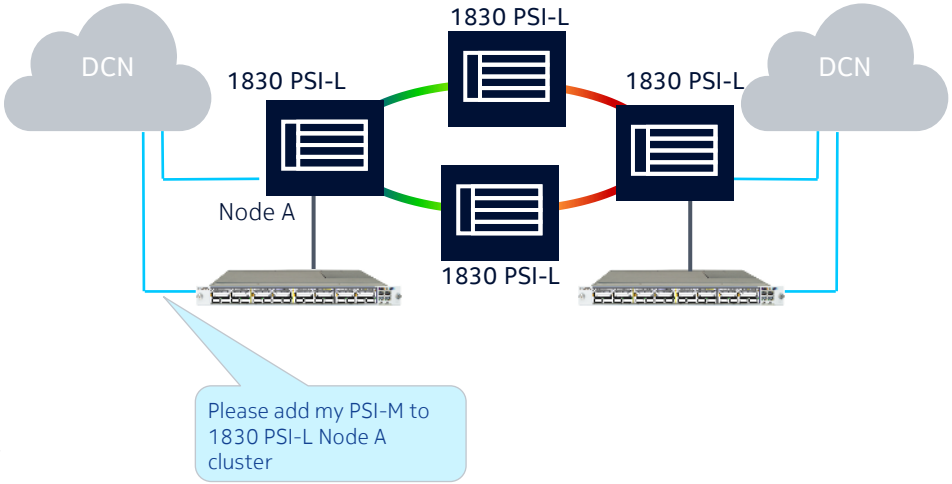
© Nokia 2023 | Confidential

Cluster Configuration

Cluster Configuration

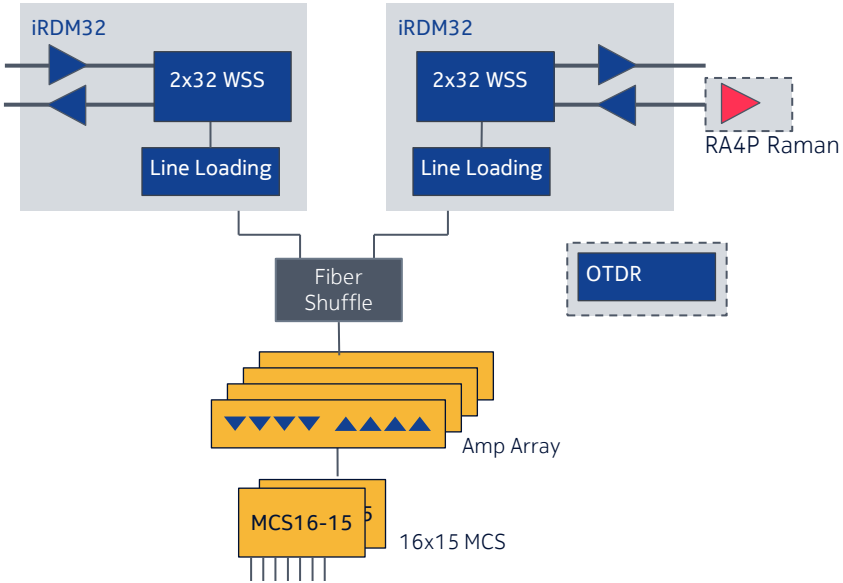
- Allows 1830 Node to control and adjust transponders (optical power levels) on subtending “cluster” shelves
  - PSI-M
  - PSS-8/16II/32

- Overview
  - Both 1830 PSI-L and PSI-M connected to local DCN (LAN),
    - allows communication between systems
  - Systems provisioned for “cluster configuration
  - Interconnecting ports & link identified as part of cluster
  - With cluster configurations established, 1830 PSI-L automatically instructs PSI-M to adjust Tx optical power levels to the target levels



© Nokia 2023 | Confidential

PSI-L ROADM Example

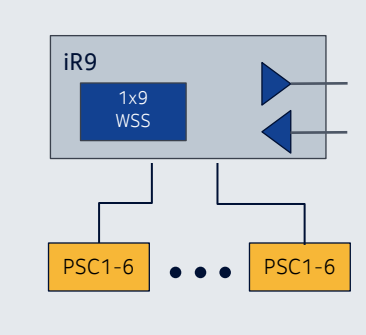


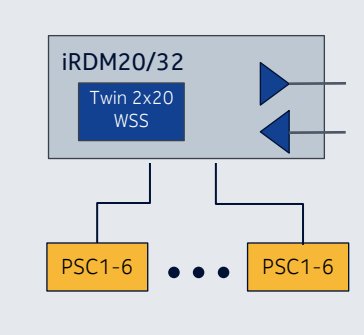


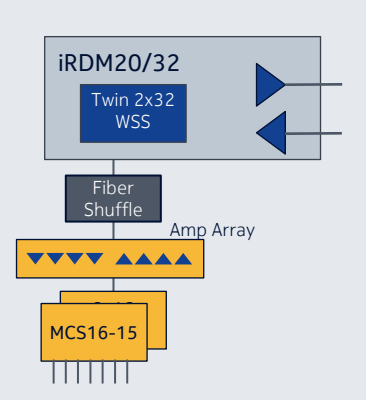
Preconfigured for 16-deg support

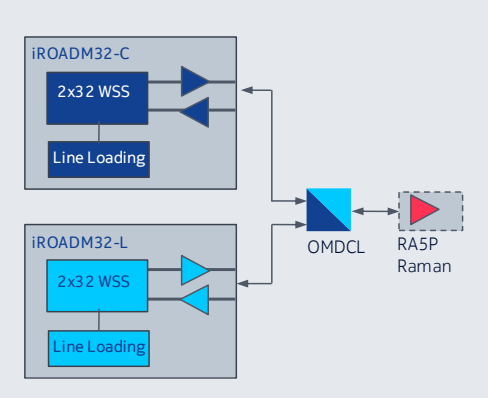
 © Nokia 2023 | Confidential

PSI-L ROADM Configurations









PSI-L Configurations

- CF ROADM
  - iR9 /20 /32
  - PSC1-6 Add/Drop
- CDC-F ROADM
  - iRDM20 / 32
  - MCS8x16 & MCS16x15
  - MxN824
- C+L ROADM
  - iRDM32 C+L CDC-F ROADM
- Pt-Pt applications
  - PSI-4L with SFD64 / SFD44 (R22.2H)

© Nokia 2023 | Confidential

Maximize critical network scalability, flexibility, and resilience with Nokia application-optimized optical line systems

Scalable

2x fiber capacity



Cost-effectively and seamlessly expand network capacity

Seamless C+L band adoption solutions (modular and integrated)

Flexible

No compromises



Improve network bandwidth management flexibility and simplify architectures and operations

Integrated ROADM solutions optimized from edge to core, C-F and CDC-F

Resilient

Maximum uptime



Maximize service uptime and optimize network resource utilization

Proven network protection and restoration options

© 2024 Nokia

Backup

 © Nokia 2023

1830 PSS-16II High Level Details



1830 PSS-16II

1830 PSS-16	FEATURE HIGHLIGHTS
CHASSIS	• Up to 3.2T of capacity in 8 RU • 200G/slot, 200G slot-to-slot interconnectivity • Distributed blade based switching (ODUk, Packet)
INTERFACES	• 8 full-height / 16 half-height slots for various service or line cards • Wide choice of configuration options (Pt – Pt FOADM, classic ROADM, CDC-F ROADM, C-F ROADM) • 96 channel WDM • Client: Sub-10G, 10G, 40G, 100G, 400GE • Line: 10G to +600G DWDM
SERVICES	• Ethernet, OTN, Data Center, and SONET/SDH cards
NETWORKING	• Point-to-point, Ring, Mesh • Transparent/groomed services • OTN/Packet aggregation
POWER	• Deployment in CO, Hut/CEV (ILA) • DC or AC power

© Nokia 2023 | Confidential

1830 PSS-8 High Level Details



1830 PSS-8

1830 PSS-8	FEATURE HIGHLIGHTS
CHASSIS	• Up to 1.6T of capacity in a 3 RU chassis • 200G/slot, 200G slot-to-slot interconnectivity • Distributed blade based switching (ODUk, Packet)
INTERFACES	• 4 full-height / 8 half-height slots for various service or line cards • Wide choice of configuration options (Pt – Pt FOADM, classic ROADM, CDC-F ROADM, C-F ROADM) • 96 channel WDM • Client: Sub-10G, 10G, 40G, 100G, 400GE • Line: 10G to +600G DWDM
SERVICES	• Ethernet, OTN, Data Center, and SONET/SDH cards
NETWORKING	• Point-to-point, Ring, Mesh • Transparent/groomed services • OTN/Packet aggregation
POWER	• Deployment in CO, ILA (Hut/CEV), or OSP environment • DC or AC power

1830 PSI-L High Level Details



PSI-L 8-slot



PSI-L 4-slot

1830 PSI-L	FEATURE HIGHLIGHTS
CHASSIS	• PSI-L Photonic Service Interface - Line • New, DCI optimized form factor • Front / Back airflow • 435mm depth – 19”/23”/ETSI rack or DCI cabinet mount • PSI-L designed WDM Line & Add/Drop units
INTERFACES	• 8 full-height slots for various WDM line cards • Supports CDC-F and C-F ROADM configurations • Supports C+L Band for double capacity • Optimized for DCI applications • Support for PSI-M / 1830 Transponders in subtending “cluster” shelf
SERVICES	• PSI-M / 1830 Transponders via cluster configuration
NETWORKING	• Point-to-point, Ring, Mesh • DCI regional / LH networks • Open line systems
POWER	• Deployment in CO environment • DC or AC power

 © Nokia 2023 | Confidential


NOKIA

Copyright and confidentiality

The contents of this document are proprietary and confidential property of Nokia. This document is provided subject to confidentiality obligations of the applicable agreement(s).

This document is intended for use by Nokia's customers and collaborators only for the purpose for which this document is submitted by Nokia. No part of this document may be reproduced or made available to the public or to any third party in any form or means without the prior written permission of Nokia. This document is to be used by properly trained professional personnel. Any use of the contents in this document is limited strictly to the use(s) specifically created in the applicable agreement(s) under which the document is submitted. The user of this document may voluntarily provide suggestions, comments or other feedback to Nokia in respect of the contents of this document ("Feedback").

Such Feedback may be used in Nokia products and related specifications or other documentation. Accordingly, if the user of this document gives Nokia Feedback on the contents of this document, Nokia may freely use, disclose, reproduce, license, distribute and otherwise commercialize the feedback in any Nokia product, technology, service, specification or other documentation.

Nokia operates a policy of ongoing development. Nokia reserves the right to make changes and improvements to any of the products and/or services described in this document or withdraw this document at any time without prior notice.

The contents of this document are provided "as is". Except as required by applicable law, no warranties of any kind, either express or implied, including, but not limited to, the implied warranties of merchantability and fitness for a particular purpose, are made in relation to the accuracy, reliability or contents of this document. NOKIA SHALL NOT BE RESPONSIBLE IN ANY EVENT FOR ERRORS IN THIS DOCUMENT or for any loss of data or income or any special, incidental, consequential, indirect or direct damages howsoever caused, that might arise from the use of this document or any contents of this document.

This document and the product(s) it describes are protected by copyright according to the applicable laws.

Nokia is a registered trademark of Nokia Corporation. Other product and company names mentioned herein may be trademarks or trade names of their respective owners.